Context Reduces Need for Action Chunking with Non-Markovian Policies

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Abstract

In the context of imitation learning, the benefits of action chunking, the practice of recalculating inputs to a dynamical system only every ℓ time steps, have been analyzed and shown to be beneficial for Markovian experts and learned policies via a stability-theoretic framework. We extend this framework to the case of non-Markovian experts and policies to provide theoretical guarantees validating the empirical observations of **?** that longer context lengths paired with shorter action chunk lengths lead to increased performance in robotic systems. We do this by modeling the dynamical system as a partially observable one and analyzing the problem from this viewpoint. We perform robotic simulations to validate our theoretical results and provide finer insights.

1. Introduction and Related Works

Learning behaviors from an expert demonstration is known as imitation learning (IL), and has been in use from the early days of autonomous robotic systems to the modern day [**?**, **?**]. The development of imitation learning techniques has been accompanied by many algorithmic interventions that are, according to innumerable experiments, critical for good performance of learned policies [**?**]. One such intervention, and the focus of this project, is action chunking, where a learned policy is only queried every ℓ time steps, and uses a nominal dynamics model to generate the system inputs for the next ℓ time steps. **?** posit that the success of action chunking is not due to partial observability, as it still has benefits in fully-observable control, but rather is due to stability theoretic reasons. They further provide theoretical guarantees on how successful action chunking is assuming a Markovian expert, policy, and stable systems. Complementing this, **?** show how stabilizing feedback controllers enable imitation of non-Markovian demonstrations. Our contribution will be to extend the framework of **?** to one without full state observation to justify when longer observation histories allow a learned policy to execute shorter action chunks.

2. Problem Formulation

We consider a discrete-time dynamical system with output

$$x_+ = f(x, u), \quad y = \psi(x) \tag{DS}$$

where $x_t \in \mathbb{R}^n, y_t \in \mathbb{R}^p, u_t \in \mathbb{R}^m$. Given trajectories $\{(x_t^{*,(i)}, y_t^{*,(i)}, u_t^{*,(i)})\}$ generated by an expert policy $u_t^* = \pi^*(x_t^*)$, a learned policy $\hat{\pi}$ can be trained to imitate the expert. We investigate the setting where the learned policy $\hat{\pi}$ is a function of the inputs and outputs over the past h time steps. More precisely, deployment of the learned policy gives a trajectory $(\hat{y}_t, \hat{u}_t)$ where $\hat{u}_t =$

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$\hat{\pi}(\hat{y}_t, ..., \hat{y}_{t-h+1}, \hat{u}_{t-1}, ..., \hat{u}_{t-h+1})$. To shorten notation, we denote $z_t = (y_t, ..., y_{t-h+1}, u_{t-1}, ..., u_{t-h+1})$ with the respective stars and hats as necessary.

In this project, we seek to theoretically ground the empirical observation and intuitive idea that increasing the context length h decreases the minimum required chunk length ℓ for an action-chunked policy to ensure stability of the closed-loop system.

3. Current Progress

Our progress so far consists of a proposition and corollary in the style of **?**, simulations further corroborating the observations of **?**, and a clear direction of where we need to go from here. We will now set up and state our main theoretical results so far.

Consider (DS) and two trajectories of it (x^1, u^1, y^1), (x^2, u^2, y^2). Define $\delta y_t = y_t^2 - y_t^1$ and similarly define $\delta x, \delta u, \delta z$. Assume then that the system satisfies

$$||\delta y_{t+k}|| \leq C\rho^k ||\delta z_t||_{\mathcal{Z}} + C\sum_{j<k} \rho^{k-1-j} ||\delta u_{t+j}|| \tag{EδIOS}$$

for all t, j for some $C \geq 1$, $\rho \in (0, 1)$. (We are aware that this perhaps isn't exactly the natural definition of incremental input-to-output stability, but for the sake of not coming up with weird names, we call this exponential incremental input-to-output stability in this work.) This condition is analogous to the EδISS condition used extensively in **?**, except now considers stability in the output relative to the context. This condition is approximately as weak as EδISS, which is to say, relatively weak for many systems of interest, such as robotic arms. In particular, EδIOS reduces exactly to EδISS when $\psi = \text{id}$ and $h = 1$. For ease of computations later, the norm of context vectors will always, even when not explicitly notated, be the sum of the Euclidean norms of the block entries, that is,

$$||z_t||_{\mathcal{Z}} = ||y_t|| + \cdots + ||y_{t-h+1}|| + ||u_{t-1}|| + \cdots + ||u_{t-h+1}||$$

Note that $||z_t||_2 \leq ||z_t||_{\mathcal{Z}}$, where $||\cdot||_2$ denotes the Euclidean norm. Thus, any D-Lipschitz (in the Euclidean norm) function that takes a context vector as input is also D-Lipschitz with respect to the $\mathcal{Z}$ norm.

Take the class of chunked policies $[\pi] = (\pi_0, ..., \pi_{\ell-1})$ with $\pi_k : \mathcal{Z} \to \mathbb{R}^m$, as

$$\Pi_\ell(D, \rho) = \{[\pi] : \pi_k(z) = c_k + \rho^k g_k(z), \text{ where } g_k \text{ is } D\text{-Lipschitz}\}$$

Then deploy the policy π in the typical chunked policy fashion as $u_{t_j+k} = \pi_k(z_{t_j}) + v_{t_j+k}$ with $t_j = j\ell$. By construction, $\pi \in \Pi_\ell(D, \rho)$ has $||\pi_k(z') - \pi_k(z)|| \leq D\rho^k ||z' - z||_{\mathcal{Z}}$.

We stand on the shoulders of giants in proving the following proposition, whose proof we surreptitiously relegate to the appendix:

Proposition 1 *Fix some $\alpha \in (0, 1)$ and $D \geq 0$. Define $\bar{\rho} = \rho^{1-\alpha}$, $C_h = 2hC\rho^{-(h-1)}$, $\bar{C} = C\left(1 + \frac{D}{ea\rho \log(1/\rho)}\right)$, and $\tilde{C} = \bar{C}C_h$. If (EδIOS) holds, $\ell \geq h - 1$, and $C_h\left(1 + \frac{D\ell}{\rho}\right)\rho^{a\ell} \leq 1$, then for every $\pi \in \Pi_\ell(D, \rho)$, f^π is such that*

$$||\delta y_t|| \leq \bar{C}\bar{\rho}^t ||\delta z_0||_{\mathcal{Z}} + \tilde{C}\sum_{j<t} \bar{\rho}^{t-1-j} ||\delta v_j||$$

This proposition gives that the closed loop system under an appropriate chunked policy with long enough chunk length is also EδIOS. Given an expert trajectory (y_t^*, u_t^*) of length T where the context z_t^* is defined as usual, and a learned policy $\hat{\pi}$ that generates a trajectory $(\hat{y}_t, \hat{\pi}(z_{t_j}))$, where the expert and learned policies are rolled out on systems with the same initial state. Define the rollout (trajectory) loss and training (demo) loss respectively as

$$J_{\text{traj}}(\hat{\pi}) = \sum_{t=0}^{T-1} ||\hat{y}_t - y_t^*||, \quad J_{\text{demo}} = \sum_{t=0}^{T-1} ||\hat{\pi}(z_{t_j}) - u_t^*||$$

Then the next result follows:

Corollary 1 *If (EδIOS) holds, for $\hat{\pi} \in \Pi_\ell(D, \rho)$ and $\ell \geq \ell_{min}$, we have for every horizon T that*

$$J_{traj}(\hat{\pi}) \leq \frac{\tilde{C}}{1-\bar{\rho}} J_{demo}(\hat{\pi})$$

4. Future Work

Obviously we have not directly attacked our ultimate goal yet. Rather, we have begun setting theoretical foundations that will hopefully aid us in achieving it. It remains to be proven that the chunk length can be shortened with increasing context length. Indeed, Proposition 1 requires a chunk length that is greater than the context length, nearly the opposite of what we want to prove! We will need to establish an upper bound on chunk lengths that are sufficient to still ensure closed loop stability. It makes intuitive sense that such upper bounds should exist, as in physical examples you don't run an infinite-length chunk, but we have yet to establish such bounds. Once such a bound is established, we can utilize it in our attack on the chunk length by increasing the horizon length, which will hopefully be a sufficient tool to reduce the upper bound.

There are several other housekeeping matters for us to attend to. First, we will run more comprehensive and digestible simulations for our final report. Additionally, it remains to be proven that certain classes of systems are actually EδIOS, such as stable linear systems. Proving that classes of systems are indeed EδIOS will lend further credence to the value of our theoretical work.

5. Simulations: experiment, data, and results

The experiment. We use **?** 'hard example' at $\mu = 1/8$ as described in proposition 4.1. The expert sees x_1, x_2, while the learner sees only $y = x_2$. Two observations and the action between them suffice to reconstruct the state from the system equations ($h = 2$). We compare action sequences and single actions, each with and without history. History lengths are $r = 2, 4, 8, 16$; no history meaning a markovian learner. A sequence predicts $H = 16$ actions and executes $\ell = 2, 4, 8, 16$ before observing again. Single-action control uses $H = \ell = 1$.

The data and learner. Narrow data contain expert trajectories that stay on $x_1 = 0$, so no correction of x_1 is demonstrated. Recovery data add small random changes to the expert's applied actions, $u = K_1 x + 0.05\xi$, $\xi \sim \mathcal{N}(0, I)$. At each visited state, the training data is the action sequence the expert would apply without further random changes, $K_1 F^j x$. Inputs retain the observations and actions actually applied. This adapts noisy collection and expert correction (**??**), motivated by **?**.

We use ridge regression to obtain a learned controller. It minimizes action-prediction error plus a coefficient penalty ($\lambda = 10^{-6}$). Three seeds (818–820) each generate 32 trajectories of 80 steps per dataset type. Training target times match across history and prediction length. The 60 fitted models produce 150 four-condition evaluations and 30 checks with $H = 16, \ell = 1$. Every check matches the separately trained single-action learner. At fixed history, varying execution uses the same fitted predictor.

For testing, we fill history with 15 expert steps, then introduce a signed change of magnitude 0.0005–0.0015 in hidden x_1. All learners share 20 new starting states with x_2 uniform on $[-1, 1]$. Each controls for 200 steps. An expert rollout from the same state supplies the reference. It is then evaluated on MSE from the expert. We retain unstable losses and check stability separately from exact linear feedback dynamics. Matched states without the change and separate 1000-step examples are also saved.

Training	History r	Prediction horizon H	Execution horizon ℓ	Test MSE	Stable?
Narrow	1	16	16	5.19×10^{-3}	No
Narrow	8	1	1	7.58×10^{-4}	No
Narrow	8	16	8	2.32×10^{-6}	Yes
Narrow	16	1	1	6.65×10^{-7}	Yes
Recovery	1	16	16	2.53×10^{-1}	No
Recovery	2	1	1	7.33×10^{-9}	Yes
Recovery	2	16	16	2.71×10^{-7}	Yes

Selected settings; all three seeds agree on stability in every tested setting.

The results. Without history, every fitted learner remains unstable under both datasets. Longer history reduce short-term error without fixing this: narrow-data execution sixteen has MSE 0.00519

With recovery data, history two and single-action feedback nearly match the expert (MSE 7.334×10^{-9}). Executing sixteen actions increases error about 37-fold: the learner waits longer to respond after the initially invisible departure affects a new observation. All $r \geq 2$ settings are stable. Longer history is not always better; one-step MSE rises to 3.04×10^{-8} at $r = 16$.

With narrow data, history two is observable but all executions fail stability. History eight is stable only at execution eight or sixteen; execution eight improves MSE 327-fold over single actions. History sixteen is already stable without sequences, and execution sixteen improves error only 11%. Extra history changes the ridge-selected feedback, without supplying missing corrective examples. Thus Proposition 1's representation result does not imply successful learning.

History plus corrective examples makes single-action control effective here; observability alone does not remove the benefit of sequences. These findings concern one stable system, three seeds, small departures, expert initialization, and noiseless testing. Training and test amplitudes also differ. Neither these data nor the theory establishes universal improvement with history or execution length.

6. Appendix

6.1. Proof of Proposition 1

Proof Consider the j-th chunk, which begins at time t_j and let $t = t_j + k$. We have by definition of $\Pi_\ell(D, \rho)$ that

$$||\delta u_t|| \leq D\rho^k ||\delta z_{t_j}|| + ||\delta v_t||$$

By EδIOS, for $0 \leq k \leq \ell$

$$\begin{aligned} ||\delta y_t|| &\leq C\rho^k ||\delta z_{t_j}|| + C \sum_{i<k} \rho^{k-1-i} (D\rho^i ||z_{t_j}|| + ||\delta v_{t_j+i}||) \\ &= (C\rho^k + CDk\rho^{k-1}) ||\delta z_{t_j}|| + C \sum_{i<k} \rho^{k-1-i} ||\delta v_{t_j+i}|| \end{aligned} \tag{$*$}$$

By supposition, $\ell \geq h-1$, all samples in $z_{t_{j+1}}$ like within the j-th chunk. Note that

$$||\delta z_t|| = \sum_{i=0}^{h-1} ||\delta y_{t-i}|| + \sum_{i=1}^{h-1} ||\delta u_{t-i}||$$

So then

$$||\delta z_{t_{j+1}}|| = \sum_{k=\ell-h+1}^{\ell} ||\delta y_{t_j+k}|| + \sum_{k=\ell-h+1}^{\ell-1} ||\delta u_{t_j+k}||$$

Inserting then the inequalities we derived above,

$$\begin{aligned} ||\delta z_{t_{j+1}}|| &\leq \sum_{k=\ell-h+1}^{\ell} \left((C\rho^k + CDk\rho^{k-1}) ||\delta z_{t_j}|| + C \sum_{i<k} \rho^{k-1-i} ||\delta v_{t_j+i}|| \right) + \sum_{k=\ell-h+1}^{\ell-1} \left(D\rho^k ||\delta z_{t_j}|| + ||\delta v_{t_j+k}|| \right) \\ &= \left(\sum_{k=\ell-h+1}^{\ell} (C\rho^k + CDk\rho^{k-1}) + \sum_{k=\ell-h+1}^{\ell-1} D\rho^k \right) ||\delta z_{t_j}|| \\ &\quad + \left(C \sum_{k=\ell-h+1}^{\ell} \sum_{i<k} \rho^{k-1-i} ||\delta v_{t_j+i}|| + \sum_{k=\ell-h+1}^{\ell-1} ||\delta v_{t_j+k}|| \right) \end{aligned}$$

When $k \geq \ell - h + 1$, we have $\rho^k \leq \rho^{\ell-h+1}$ so $(C\rho^k + CDk\rho^{k-1}) \leq C\rho^{\ell-h+1}(1 + D\ell/\rho)$ because $k \leq \ell$. Thus, the first big term in parentheses in the last line above, which we will call Θ_ℓ, has

$$\begin{aligned} \Theta_\ell &\leq h\rho^{\ell-h+1} \left(C(1 + D\ell\rho) + D \right) \\ &\leq 2hC\rho^{-(h-1)} \rho^\ell (1 + D\ell/\rho) \\ &= C_h \rho^\ell \left(1 + \frac{D\ell}{\rho} \right) \end{aligned}$$

where the second inequality follows from the fact that $D \leq D\ell/\rho$ and $C \geq 1$, so $D \leq C(1 + D\ell/\rho)$. Then by the assumption in the proposition, $\Theta_\ell \leq \rho^{\ell - \alpha\ell} = \bar{\rho}^\ell$.

For the second big term in parentheses, which we call B_j, we first examine the double summation. Note that each $\rho^{k-1-i} \leq \rho^{\ell-h-i}$, and there are no more than h terms of the outer sum, so the

coefficient of each $||\delta v_{t_j+i}||$ is no more than $hC\rho^{\ell-h-i}$. For the single summation, the coefficient of $||\delta z_{t_j+k}||$ is 1 for $\ell - h + 1 \leq k \leq \ell$. Rearranging, we have for such k that $\ell - h - k \leq -1$, so $1 \leq \rho^{-(h-1)}\rho^{\ell-1-k}$, so the coefficients in the single sum are certainly less than or equal to that. Thus,

$$\begin{aligned} B_j &\leq \sum_{k=\ell-h+1}^{\ell-1} \left(hC\rho^{\ell-h-k} + \rho^{-(h-1)}\rho^{\ell-1-k}\right) ||\delta v_{t_j+k}|| \\ &\leq \sum_{k=\ell-h+1}^{\ell-1} \left(2hC\rho^{-(h-1)}\rho^{\ell-1-k}\right) ||\delta v_{t_j+k}|| \\ &= C_h \sum_{k=\ell-h+1}^{\ell-1} \rho^{\ell-1-k}||\delta v_{t_j+k}|| \end{aligned}$$

Combining the bounds on Θ_ℓ and B_j, we get that

$$\begin{aligned} ||\delta z_{t_{j+1}}|| &\leq \bar{\rho}^{-\ell}||\delta_{t_j}|| + C_h \sum_{k=\ell-h+1}^{\ell-1} \rho^{\ell-1-k}||\delta v_{t_j+k}|| \\ &\leq \bar{\rho}^{-\ell}||\delta_{t_j}|| + C_h \sum_{k=t_{j+1}-h}^{t_{j+1}-1} \rho^{t_{j+1}-k-1}||\delta v_k|| \\ &\leq \bar{\rho}^{-\ell}||\delta_{t_j}|| + C_h \sum_{k=t_j}^{t_{j+1}-1} \rho^{t_{j+1}-k-1}||\delta v_k|| \end{aligned}$$

Unrolling this recursively, we get

$$||\delta z_{t_j}|| \leq \bar{\rho}^{t_j}||\delta z_0|| + C_h \sum_{k=0}^{t_j-1} \bar{\rho}^{t_j-1-k}||\delta v_k|| \qquad (**)$$

Now we consider

$$\rho^{\alpha k}\left(1 + \frac{Dk}{\rho}\right) = \rho^{\alpha k} + \frac{D}{\rho}k\rho^{\alpha k} \leq 1 + \frac{D}{\rho}ke^{-\alpha k \log(1/\rho)} \leq 1 + \sup_{q\geq 0}\frac{D}{\rho}qe^{-\alpha q\log(1/\rho)} = 1 + \frac{D}{e\alpha\rho\log(1/\rho)}$$

Hence,

$$C\rho^k + CDk\rho^{k-1} = C\rho^k\left(1 + \frac{Dk}{\rho}\right) \leq C\rho^{\alpha k}\left(1 + \frac{Dk}{\rho}\right) \leq \bar{C}\bar{\rho}^k$$

Inserting this bound and $(**)$ into $(*)$ gives

$$\begin{aligned} ||\delta y_t|| &\leq \bar{C}\bar{\rho}^k\left(\bar{\rho}^{t_j}||\delta z_0|| + C_h\sum_{q=0}^{t_j-1}\bar{\rho}^{t_j-1-q}||\delta v_q||\right) + C\sum_{i<k}\rho^{k-1-i}||\delta v_{t_j+i}|| \\ &\leq \bar{C}\bar{\rho}^{-t}||\delta z_0|| + \bar{C}C_h\sum_{q=0}^{t_j-1}\bar{\rho}^{t_j+k-1-q}||\delta v_q|| + CC_h\sum_{i=0}^{k-1}\bar{\rho}^{k-1-i}||\delta v_{t_j+i}|| \\ &\leq \bar{C}\bar{\rho}^{-t}||\delta z_0|| + \tilde{C}\sum_{q=0}^{t}\bar{\rho}^{t-1-q}||\delta v_q|| \end{aligned}$$

where the second line follows because $C_h \geq 1$ and $\rho \leq \bar{\rho}$.

■

6.2. Proof of Corollary 1

Proof Assume $\hat{z}_0 = z_0^*$. Then the previous proposition gives that

$$||\hat{y}_t - y_t^*|| \leq \tilde{C} \sum_{q=0}^{t} \bar{\rho}^{t-1-q} ||\hat{\pi}(\hat{z}_q) - u_q^*||$$

Summing over $t = 1, ..., T$ gives

$$\sum_{t=1}^{T} ||\hat{y}_t - y_t^*|| \leq \tilde{C} \sum_{t=1}^{T} \sum_{q=0}^{t-1} \bar{\rho}^{t-1-q} ||\hat{\pi}(z_{t_j}^*) - u_t^*|| \leq \frac{\tilde{C}}{1-\bar{\rho}} \sum_{t=0}^{T-1} ||\hat{\pi}(z_{t_j}^*) - u_t^*||$$

■